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1. SAMPLING :

- Low spatial resolution means fewer pixels are used to represent the image.
- The problem statement is insufficient spatial sampling.
- Therefore small images/objects appear unclear. By increasing the sampling rate or using a high-resolution camera can improve the image details.

2. DARK AND NOISY IMAGE :

- Can use histogram Equalization to improve brightness and contrast.
- A median filter can reduce noise.
- Therefore this makes the right image clearer.

3. QUANTIZATION :

- ⇒ Quantization converts continuous intensity values to discrete digital values.
- ⇒ For EXAMPLE : 8 bit image has 256 intensity values (0-255).
- ⇒ Therefore it helps in digital image storage and processing.

4. UNEVEN LIGHTING:

- ⇒ use contrast enhancement or CLAHE to improve the image.
- ⇒ It increases the visibility of defects in poorly lit areas. Thus, defects become easier to detect.

5. IMAGE SEGMENTATION:

Image Segmentation separates the image into regions such as roads, vehicles, and pedestrians. It helps the car identify different objects. Therefore this supports safer autonomous driving.

6. EDGE DETECTION:

- ⇒ Use Canny edge detection to detect sudden intensity changes.
- ⇒ Cracks appear as edges in the image. Therefore this helps identify the location and shape of defects.

7. FILTERING:

- ⇒ use a median filter to remove unwanted noise. It reduces noise while preserving important edges.
- ⇒ This makes the tumor easier to identify.

8.

OBJECT DETECTION :

- ⇒ Object detection identifies vehicles and their locations.
- ⇒ Tracking follows the vehicles across video frames. it helps monitor traffic and detect congestion.

9.

MOTION BLUR :

- ⇒ The blur is caused by object movement during image capture.
- ⇒ use image / deblurring / restoration to improve the image.
- ⇒ A faster shutter speed can also reduce motion blur.

10.

IMAGE CLASSIFICATION :

- ⇒ Image classification separates products into good and defective categories.
- ⇒ A trained model identifies defects such as cracks or scratches.
- ⇒ This supports fast & automatic quality inspection.